



Thermo-fluid coupled model based on laser energy deposition for high-repetition rate ultrafast laser welding of glass and metal



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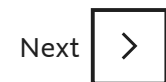
Highlights

- A thermo-fluid model is built for high-repetition rate ultrafast laser welding of glass and metal.
- 2D steady-state single-pulse model is built achieving 8-order efficiency gain over 3D transient.

- Non-uniform energy deposition within the glass at high repetition rate is analyzed.
- The flow at the glass-metal interface primarily driven by the recoil pressure is studied.

Abstract

Due to its unique threshold effect, ultrafast laser has significant advantages in glass welding. However, the mechanism of non-uniform energy deposition during the interaction between high-repetition rate laser and glass is not well understood. A thermo-fluid coupled model based on laser energy deposition is proposed to analyze the process of high-repetition rate ultrafast laser welding of glass and metal. By reducing the single-pulse model to a two-dimensional steady-state, computational efficiency can be greatly improved, allowing for the solution of each single pulse in the multi-pulse model to show the differences in laser energy deposition. The model indicates that the laser energy deposition area periodically moves from the glass-metal interface into the interior of the glass. Subsequently, the thermal and fluid response of the material after laser energy deposition is analyzed, providing theoretical guidance for optimizing the process parameters of ultrafast laser glass-metal welding.



Keywords

Ultrafast laser; Glass-metal welding; Multiple pulses; Thermo-fluid coupled model; High-repetition rate

1. Introduction

Ultrafast lasers have pulse widths in the femtosecond to picosecond range. Their short pulse width, high peak power, and threshold effect make them highly advantageous in material processing [[1], [2], [3]], offering high precision, low heat impact, and broad material applicability. In particular, the threshold effect of ultrafast lasers gives them unique advantages in processing transparent materials such as glass [4,5]. They can maintain the transparency of glass in low-intensity areas while absorbing laser energy in the high-intensity focal region. This allows ultrafast lasers to selectively deposit energy within the glass. As the deposited energy increases from low to high, the glass can be modified for applications such as optical waveguide manufacturing [6], micro-welding [7,8], or ablation applications like drilling and surface texturing [9].

Ultrafast laser glass welding deposits laser energy onto the interface of the glasses to be welded, causing the glass to melt and form a molten pool, thereby connecting. In 2005, Tamaki et al. [10] from Osaka University in Japan used near-infrared femtosecond laser to weld two pieces of silica glass, marking the first report of ultrafast lasers being applied to glass welding. Early ultrafast laser glass welding used low laser repetition rates and slow scanning speeds, resulting in small welding pools, low efficiency, and high requirements for the surface quality of the glass and the focus of the laser. Watanabe et al. [11] used a femtosecond laser with a repetition rate of 1 kHz and a scanning speed of 0.1–1 mm/s to weld borosilicate glass and fused quartz, requiring the glass to be in optical contact before welding (maximum permissible gap $< \lambda/4$, where λ is the wavelength of light). With the development of ultrafast laser glass welding technology, high repetition rates and fast scanning speeds have gradually become favored by researchers for their advantages of larger molten pools, higher efficiency, and the ability to bridge larger gaps. High repetition rates (> 200 kHz) mean that the pulse period is shorter than the time required for heat to diffuse from the focal volume, thus leading to heat accumulation between pulses to expand the molten pool. Chen et al. [12] used a picosecond laser with a repetition rate of 400 kHz to weld glass without the requirement for pre-existing optical contact. Richter et al. [13] used a burst-mode femtosecond laser with individual pulse separation of 20 ns within one burst to form a large molten pool with $450 \mu\text{m} \times 160 \mu\text{m}$, thereby successfully welding glass with a gap of $3 \mu\text{m}$. Chen et al. [14] utilized a picosecond laser with a small-scale rapid oscillating scan (1 MHz, 1000 mm/s) to achieve the welding of glasses with natural stacking contact (gap $\approx 10 \mu\text{m}$).

Although high repetition rates have become mainstream in ultrafast laser glass welding, the mechanism of energy deposition within the glass by high repetition rate lasers is still not clear. High repetition rate ultrafast laser scanning can form non-uniform structures along the laser scanning direction within the glass, indicating that the laser energy deposition is non-uniform. Bellouard et al. [15] used a femtosecond laser with a high repetition rate of 9.4 MHz to irradiate fused quartz and demonstrated from a top view some periodically arranged teardrop-shaped modified structures and regularly arranged self-organized bubbles within these structures. Kissi et al. [16] used a femtosecond laser with a repetition rate of 3 MHz to form a series of bubble

structures or self-organized nano-grating structures within the glass under different laser energy inputs. This demonstrates the complexity of the interaction between high repetition rate lasers and glass.

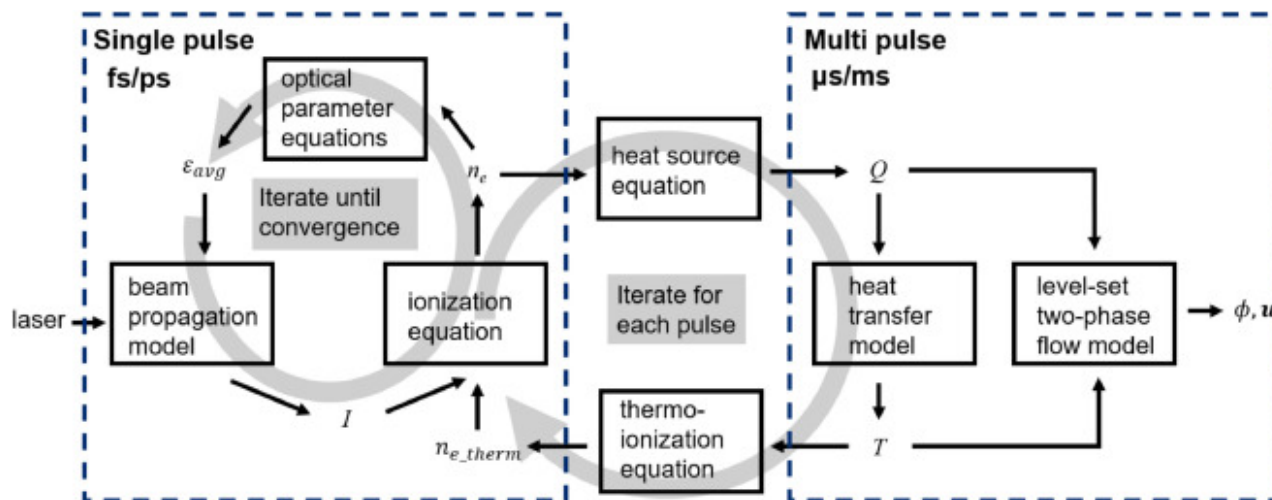
Existing simulation studies mostly focus on the interaction of a single laser pulse with glass [17,18], or the shielding effect produced by several pulses acting in an extremely short time (on the picosecond scale) [19]. A few studies have performed simulations of multiple pulses [20,21], but these are based on a simple superposition of single-pulse heat sources, which can reflect the thermal accumulation effect under multiple pulse action, and cannot reflect the non-uniform deposition phenomenon of high repetition rate lasers interacting with glass. Miyamoto et al. [22] observed with a high-speed camera that under the action of high repetition rate lasers, high absorption small plasmas periodically move towards the laser source near the geometric focus, thereby forming non-uniform structures within the glass. They further realized the simulation of non-uniform energy deposition of high repetition rate lasers by introducing the mechanism of thermal ionization. Their simulation results are qualitatively consistent with the observed plasma motion, but the overly simplified simulation settings, such as a one-dimensional single-pulse model and analytical solutions of the temperature field that limit the glass thermal properties to constant values, resulting in the simulated periodic plasma motion having a period two orders of magnitude smaller than the experimentally observed period.

Ultrafast laser welding of glass and metal involves the absorption of laser energy by both glass and metal, as well as the transfer of energy between them, making the energy deposition process more complex than that in ultrafast laser glass welding. However, similar to ultrafast laser glass welding, non-uniform energy deposition can still be observed in ultrafast laser glass-metal welding at high repetition rates. Tian et al. [23] used a picosecond laser with a repetition rate of 400 kHz for welding glass and metal, observing periodically arranged dendritic modified structures along the scanning direction on the glass side, which were confirmed by a high-speed camera to be produced by the periodic movement of plasma towards the laser source. Current research on ultrafast laser glass-metal welding mostly focuses on parametric experiments [24] and weld seam observation [25], with no related simulation studies available yet. The mechanism of energy deposition in high repetition rate laser glass-metal welding is also currently a blank area.

In this paper, a simulation model is established to describe the process of non-uniform energy deposition in glass-metal welding at high repetition rates. Based on this, the thermal and fluid responses after laser energy deposition are analyzed, and the temperature field and the flow field at the interface by multi-pulse with the high repetition rate during the glass-metal welding process are obtained. Finally, the model is validated through experiments.

2. Numerical modeling

The numerical model for the entire ultrafast laser glass-metal welding is divided into three parts: the single-pulse section, the multi-pulse temperature field section, and the multi-pulse flow field section. The single-pulse section describes the interaction process between a single laser pulse and glass on the picosecond timescale, forming a heat source of a single pulse. The multi-pulse temperature field section describes the temperature field formed across the entire glass-metal area by the superposition of single-pulse heat sources on the microsecond timescale, and feeds back the temperature field to the single-pulse section to influence the formation of subsequent single-pulse heat sources. The single-pulse section and the multi-pulse temperature field section are solved sequentially through iterative data exchange rather than unified computation. Each applied laser pulse triggers a sequential process: first resolving the instantaneous single-pulse heat source with sub-picosecond temporal resolution, followed by simulating the cumulative temperature field evolution across the pulse sequence using microsecond-scale discretization. The multi-pulse flow field section describes the material phase change and flow process near the glass-metal interface driven by temperature on the microsecond timescale. The overall simulation framework is shown in Fig. 1.



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Fig. 1. The modelling flowchart from single-pulse to multi-pulse.

2.1. Single-pulse modeling

The single-pulse simulation consists of three parts: the beam propagation model, the ionization equation, and the optical parameter equations. The beam propagation model describes how the laser intensity is distributed as the laser propagates through the glass, and then passes this intensity distribution to the ionization equation. The ionization equation describes how the glass ionizes to produce free electrons under a certain laser intensity, essentially converting the energy of the laser into the energy of the electrons within the glass. The optical parameter equations describe how the ionized free electrons affect the optical properties of the glass, and then pass these optical parameters to the beam propagation model to influence the propagation of the laser beam. In summary, these three parts are coupled together, forming the interaction process between the laser and the glass under a single pulse.

2.1.1. Dimensionality reduction

Due to the non-uniform deposition of laser energy at high repetition rates, it indicates that the action process of each individual pulse is distinct, hence it is necessary to calculate the action process of each single pulse separately, rather than just calculating the action process of a single pulse once and then simply repeating the superposition. This requires improving the computational efficiency of the single-pulse simulation.

In this study, dimensionality reduction of the spatial and temporal scales is applied to the single-pulse simulation to enhance the computational efficiency of the single-pulse model. The dimensionality reduction in the temporal scale refers to simplifying the conventional transient simulation to a steady-state simulation. The approach of transient simulation [17] involves inputting a Gaussian temporal distribution laser into the beam propagation model, then inputting the obtained current moment's intensity into the ionization equation to get the current moment's free electron density, and subsequently obtaining the next moment's optical parameters of the glass through the optical parameter equations, which are then input into the beam propagation model to calculate the next moment's intensity. In this way, by continuously iterating the calculations, one can obtain the temporal variation process of the intensity and free electron density within the glass. Since the goal of the single-pulse simulation in this study is to form a single-pulse heat source without focusing on the changes in physical quantities during the single-pulse input process, steady-state simulation can be conducted. The approach of steady-state simulation is to input the laser into the beam propagation model at its peak power to obtain a steady-state intensity. This intensity is then extended in the time domain, assuming that the glass will experience a Gaussian temporal distribution of intensity. This intensity sequence is input into the ionization equation to derive a sequence of free electron densities. The average free electron density is then input into the optical

parameter equations to calculate the average optical parameters, which are subsequently fed back into the beam propagation model for iterative calculations until a stable value is reached.

The dimensionality reduction in spatial scale is to simplify the conventional three-dimensional (3D) simulation to a two-dimensional (2D) simulation. Since the laser, as an electromagnetic wave, has a fixed direction of the electric field, it is not possible to equivalently reduce the beam propagation model to a two-dimensional axisymmetric model through rotational symmetry (cylindrical coordinates); thus, it is simplified to a two-dimensional model (Cartesian coordinates) to preserve vector field characteristics of the electric field. However, there is a difference in the spatial distribution of Gaussian laser intensity between the 2D and 3D models. A correction factor (<1) must be applied to the intensity obtained from the 2D model to match the intensity of the 3D model.

For the 3D transient single-pulse model, the time step is on the order of 0.01 fs, and the grid size must be smaller than one-fifth of the wavelength of light, at the 100 nm scale. In contrast, the reduced 2D steady-state single-pulse model only needs to calculate the optical field under the input of the laser at peak power, and the grid size only needs to resolve the laser beam waist, which is at the micrometer scale. If the computational workload is measured by the product of the number of time steps and the number of grids, for a simulation with a laser pulse duration of 400 fs and a computational domain size of $15\text{ }\mu\text{m} \times 15\text{ }\mu\text{m} \times 70\text{ }\mu\text{m}$, the computational workload of the 2D steady-state model is reduced by eight orders of magnitude compared to the 3D transient model. For laser pulse durations on the order of picoseconds and computational domain scales on the order of millimeters, the improvement in computational efficiency of the 2D steady-state model would be even more significant.

2.1.2. Governing equations

The 2D steady-state beam propagation model is governed by the Helmholtz equation, which is a simplification of Maxwell's equations based on the slowly varying envelope and paraxial approximation assumptions, expressed by [Eq. \(1\)](#).

$$2jk_0 \frac{\partial E}{\partial z} = \Delta_{\perp} E + k_0^2 (\epsilon_{avg} - 1) E \quad (1)$$

where $k_0 = 2\pi/\lambda$ is the wave number, λ is the laser wavelength, j is the imaginary unit, $\Delta_{\perp} = \partial^2/\partial x^2$ is transverse Laplacian, E is envelope of the electric field in the z direction, ϵ_{avg} is complex relative permittivity which is the average of relative permittivity because of steady analysis. The wave vector of input laser pulse points to $-z$ direction and the pulse is polarized in the x direction.

The intensity of light I is directly proportional to the square of the electric field strength E , as expressed by Eq. (2).

$$I = \frac{1}{2} \epsilon_0 c E^2 \quad (2)$$

where ϵ_0 is the vacuum dielectric permittivity, c is the velocity of light. The correction is applied to the 2D light intensity, and the corrected light intensity I_{cor} is expressed by Eq. (3).

$$I_{cor} = \frac{\omega_0}{\omega} I \quad (3)$$

where ω_0 is the beam waist radius of the laser, and $\omega = \omega_0 \sqrt{1 + \left(\frac{z}{R_z}\right)^2}$ is beam radius of the laser along z axis, and $R_z = \pi \omega_0^2 / \lambda$ is the Rayleigh length of the laser.

The correction factor $\frac{\omega_0}{\omega}$ originates from the difference between the distribution of a 3D Gaussian laser in space, $I_{3D}(r, z) = I_0 \frac{\omega_0^2}{\omega^2} \exp\left(-\frac{2r^2}{\omega^2}\right)$, and that of a 2D Gaussian laser in space, $I_{2D}(x, z) = I_0 \frac{\omega_0}{\omega} \exp\left(-\frac{2x^2}{\omega^2}\right)$, under the condition that the central light intensity at the focus I_0 and the beam waist radius ω_0 are consistent. So, the correction factor compensates for the missing y -dimension energy distribution in the 2D formulation. It can be observed that the light intensity at the focal plane remains consistent before and after correction, while the light intensity in areas away from the laser focal plane requires a smaller correction factor to achieve equivalence from 2D to 3D.

In glass, electrons in the valence band can be excited to free electrons by strong light fields through multi-photon ionization. Subsequently, the free electrons are heated to higher energy levels by inverse bremstrahlung absorption and exchange energy with other electrons through collisions to initiate avalanche ionization. The free electrons also lose energy due to relaxation and return to the valence band. This process is described by the ionization equation Eq. (4) [26].

$$\frac{\partial n_e(t)}{\partial t} = \frac{n_v - n_e}{\tau_1} \left(a I(t) n_e(t) + \delta_N (I(t))^N \right) - \frac{n_e}{\tau_1} \quad (4)$$

where the term on the left side is the overall rate of increase in free electron density n_e , while the first term on the right side is the avalanche ionization rate and multiphoton ionization rate, which is limited by the density of excitable electrons n_v in the valence band. The second term on the right side is the decay due to relaxation of free electrons. N is the order of multiphoton

ionization, a and δ_N are corresponding coefficients for avalanche ionization and multiphoton ionization respectively, τ_1 is electron relaxation time.

The light intensity time sequence $I(t)$ is the previously obtained corrected light intensity I_{cor} expanded in the time domain in the form of a Gaussian distribution, expressed by Eq. (5).

$$I(t) = I_{cor} \times \exp\left(- (4 \ln 2) \left(\frac{t}{t_p}\right)^2\right) \quad (5)$$

where t_p is pulse duration. This implies that each point within the glass experiences a Gaussian beam with a temporal profile matching the laser pulse duration, neglecting the influence of preceding light intensities on subsequent intensities during the irradiation process.

The ionization process describes the transfer of energy from the laser to the free electrons in the glass. The total laser input energy density is $F = \int I(t) dt = \frac{I_{cor} t_p}{2\sqrt{\ln 2/\pi}}$, and the energy density transferred to the free electrons in the glass is

$Q = 2.25 U_i n_{e_max}$ [27], where U_i is the band gap energy, and n_{e_max} is the maximum value in $n_e(t)$. This represents the sum of the band gap energy and kinetic energy of the free electrons. Therefore, the average absorption coefficient of the glass is $\alpha_{h_avg} = Q/F$.

The average relative permittivity ϵ_{avg} of the glass is described by the optical parameter equation Eq. (6) [28] and is fed back into the beam propagation model.

$$\epsilon_{avg} = n^2 - \frac{\omega_p^2}{\omega^2 + j\omega/\tau} \quad (6)$$

where n is refractive index of glass, ω is laser angular frequency, τ is damping time, $\omega_p = e\sqrt{\frac{n_{e_avg}}{m_e \epsilon_0}}$ is plasma frequency, e is the electron charge, m_e is the electron mass, n_{e_avg} is the average free electron density during the ionization process, which can be obtained through the relationship between the absorption coefficient and the relative permittivity, as described by Eq. (7).

$$\alpha_{h_avg} = \frac{2\omega \operatorname{imag}(\epsilon_{avg})}{2nc} \quad (7)$$

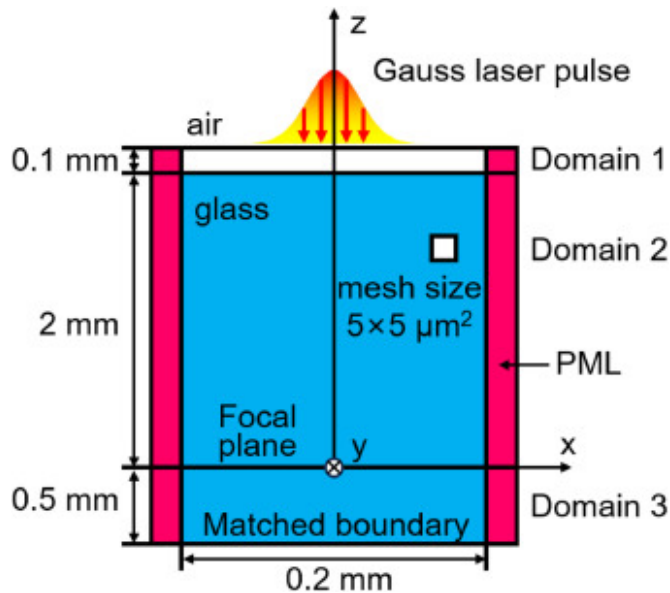
where $\operatorname{imag}(\epsilon_{avg})$ is the imaginary part of the average relative permittivity.

2.1.3. Finite element model and boundary conditions

The beam propagation model is solved using the Beam Envelope Module in COMSOL Multiphysics. The geometry of the beam propagation model consists of two layers: an upper layer of air with a thickness of 0.1 mm and a lower layer of glass with a thickness of 2.5 mm, with the focal plane set at a distance of 0.5 mm from the bottom surface, as shown in Fig. 2. The calculation domain has a width of 0.2 mm, and the grid size is $5 \mu\text{m} \times 5 \mu\text{m}$. The pink areas on the left and right sides are perfectly matched layers (PMLs), and the bottom boundary is a matching boundary, both designed to prevent laser reflection at the boundaries. The upper side is the laser input boundary, where the laser has a Gaussian spatial distribution, and the electric field intensity distribution on the boundary is described by Eq. (8).

$$E(x, z) = E_0 \sqrt{\frac{\omega_0}{\omega}} \exp\left(-\frac{x^2}{\omega^2}\right) \exp\left(-jk_0 \frac{x^2}{2R_c}\right) \exp\left(-j\left(k_0 z - \frac{\eta}{2}\right)\right) \quad (8)$$

where E_0 is the magnitude of electric field at the center of beam waist, $R_c = z + \frac{R_z^2}{z}$ is the curvature radius of wave front, $\eta = \arctan\left(\frac{z}{R_z}\right)$ is the Gouy phase-shift.



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Fig. 2. Geometry of the beam propagation model.

2.1.4. Single-pulse heat source

For ultrafast laser glass welding, domains 2 and 3 in Fig. 2 are both glass, and the energy $Q = 2.25U_i n_{e_max}$ carried by the free electrons ionized in the glass is fully deposited into the glass matrix to become a single-pulse heat source. For ultrafast laser glass-metal welding, domain 2 in Fig. 2 is glass, and domain 3 is metal. When calculating the beam propagation model, the effect of the laser reflected through the glass-metal interface on the laser intensity distribution in domain 2 is not considered, which is reasonable for the case of a rough metal surface. However, if the metal surface is smooth, the reflected laser will interfere with the incident laser, increasing the local intensity. When forming a single-pulse heat source, a volume heat source will be formed inside the glass of domain 2, consistent with glass welding, and then a surface heat source will be formed at the glass-metal interface, described by Eq. (9).

$$Q_s = (1 - R) \int I(t) dt = \frac{(1-R)I_{cor}t_p}{2\sqrt{\ln 2/\pi}} \quad (9)$$

where R is the reflectivity of the metal surface.

2.2. Multi-pulse temperature field modeling

The simulation of the multi-pulse temperature field primarily involves a heat transfer model. This model successively applies single-pulse heat sources to obtain the temperature field after multiple pulses have been applied. The temperature field then forms the thermally ionized free electron density within the glass through the thermo-ionization equation. This thermally ionized electron density is fed back into the single-pulse model, influencing the formation of the next single-pulse heat source. Each pulse undergoes such a cycle of iterative calculation.

2.2.1. Governing equations

The heat transfer model is described by the simple heat conduction equation, expressed as Eq. (10).

$$\rho c_p \frac{\partial T}{\partial t} = \nabla \cdot (k \nabla T) + q_t \quad (10)$$

where the term on the left side is the temperature rise term, ρ is the density and c_p is the specific heat capacity, while the first

term on the right side is the heat conduction term, with k being the thermal conductivity, the second term on the right side is the heat source term, which is calculated by the single-pulse model, as shown in Eq. (11).

$$q_t = \sum_{i=1}^{n_{pulse}} (Q + Q_s \delta(z)) \delta\left(t - \frac{i-1}{f}\right) \quad (11)$$

where n_{pulse} is the number of pulses, Q and Q_s are the volume heat source in the glass and the surface heat source on the metal surface, respectively, calculated from Section 2.1.4, $\delta(z)$ is Dirac delta function, which represents loading the surface heat source onto the surface at $z = 0$, $\delta\left(t - \frac{i-1}{f}\right)$ represents transient loading based on the time series with the repetition frequency f .

The thermo-ionization equation describes the relationship between temperature T and the free electron density n_{e_therm} produced by thermo-ionization, as shown in Eq. (12) [22].

$$n_{e_therm} = n_v \frac{3\sqrt{\frac{\pi}{2}} \left(\frac{k_b T}{U_i}\right)^{1.5} \exp\left(-\frac{U_i}{2k_b T}\right)}{1 + 3\sqrt{\frac{\pi}{2}} \left(\frac{k_b T}{U_i}\right)^{1.5} \exp\left(-\frac{U_i}{2k_b T}\right)} \quad (12)$$

where k_b is the Boltzmann constant. As the temperature increases, the corresponding free electron density produced by thermo-ionization also increases. Taking 3600 °C as the criterion for significant thermo-ionization [22], the corresponding free electron density due to thermo-ionization is $3e24 \text{ 1/m}^3$. Using the calculated thermo-ionization free electron density as the initial free electron density and feeding it back into the ionization equation of the single-pulse simulation implies that at high temperatures of the glass matrix, even if the light intensity is relatively low and the number of seed electrons produced by multi-photon ionization is small, thermo-ionization can still generate enough seed electrons to initiate avalanche ionization, allowing the glass to absorb the energy of the laser.

2.2.2. Finite element model and boundary conditions

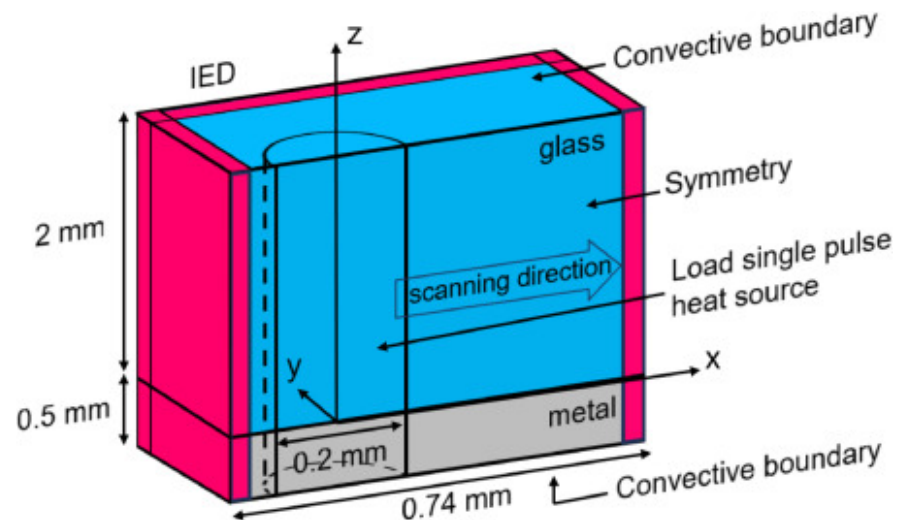
The 3D heat transfer model is solved using the Heat Transfer Module in COMSOL Multiphysics. The geometry of the heat transfer model includes two layers: an upper layer of glass with a thickness of 2 mm and a lower layer of metal with a thickness of 0.5 mm, corresponding to the single-pulse model, as shown in Fig. 3. The laser scans along the x-direction, and the model's dimension in the x-direction is 0.74 mm. The pink areas on the left, right, and back sides are infinite element domains (IEDs), approximating that these three directions have an infinitely large space where heat can freely propagate. The front side is a

symmetric boundary to reduce the scale of the model. The top and bottom sides are natural convection boundaries, with a convective heat transfer coefficient h_0 . The semi-cylinder area in the figure is the loading area for the single-pulse heat source, which is formed by rotating and mapping the 2D heat source obtained from the single-pulse model into the 3D heat transfer model. The diameter of this semi-cylinder is 0.2 mm, consistent with the width of the calculation domain in the single-pulse model. As the laser scans forward, this semi-cylinder moves in the positive x-direction at the same speed. During the welding process, it is not guaranteed that the glass and metal will be in full contact, and there may be a certain gap, so a thermal contact condition is set at their interface, and the inward heat flux at the interface is expressed by Eq. (13).

$$-n_u \cdot q_u = -h(T_d - T_u) + (1 - r)Q_b$$

$$-n_d \cdot q_d = -h(T_u - T_d) + rQ_b \quad (13)$$

where the subscript u indicates the upper interface, namely the glass interface, and the subscript d indicates the lower interface, namely the metal interface, h represents the interfacial heat transfer coefficient, characterizing the heat exchange between the glass side and the metal side of the interface, Q_b stands for the interfacial heat source, which is calculated by the single-pulse model, r is the interfacial heat distribution ratio, describing the proportion of interfacial heat source loaded onto the metal side of the interface. r is given by $r = \exp(-t^2/t_0^2)$, and as time progresses, the interfacial heat source gradually moves from the metal side to the glass side. This is used to depict the process where the absorption of laser energy by the glass interface is increased after metal spatter adheres to it under the influence of the laser. t_0 serves as a time constant for this process.



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Fig. 3. Geometry of the 3D heat transfer model.

2.3. Multi-pulse flow field modeling

The multi-pulse flow field is based on the multi-pulse temperature field calculations. To simplify the impact of the gap between the glass and metal on the interface material flow, it is assumed that when the free electron density in the glass reaches a critical value $N_{cr} = \frac{4\pi^2 c^2 m_e \epsilon_0}{\lambda^2 e^2}$, the temperature at the interface is already high enough for the glass to flow. Moreover, due to strong ionization inside the glass, a certain impact force is produced that can push the glass at the interface to fill the gap, considering at this point the glass and metal interface can reach a gap-free state. The multi-pulse temperature field and the single-pulse heat source are then input into the multi-pulse flow field model to obtain the phase field and flow field near the interface.

2.3.1. Governing equations

The multi-pulse flow field simulation is essentially a laminar level-set two-phase flow model, with one phase being glass and the other being metal. When the material temperature rises, it may undergo melting or vaporization, hence to obtain the temperature field, it is necessary to solve the heat conduction process that includes phase change. The melted material will flow

under the influence of gravity, surface tension, and vaporization recoil pressure, necessitating the solution of the Navier-Stokes equations to obtain the velocity field. To make the model tractable, the liquid in the weld pool is assumed to be laminar, incompressible and Newtonian.

During the laser heating process, the metal will melt, and its solid-liquid interface will change over time. To handle the discontinuous changes in the physical properties of the solid-liquid phases, the interface region is considered a porous medium area, which contains both liquid and solid phases, and the fraction of liquid phase f_L ($0 < f_L < 1$) is defined by the enthalpy-porosity method as Eq. (14) [29].

$$f_L = \begin{cases} 0 & T \leq T_s \\ \frac{T-T_s}{T_l-T_s} & T_s \leq T \leq T_l \\ 1 & T \geq T_l \end{cases} \quad (14)$$

where T_s and T_l are the solidus temperature and liquid temperature, respectively. $f_L = 1$ represents the liquid phase, $f_L = 0$ represents the solid phase, and $f_L = 0.5$ is considered to represent the interface between the two phases.

The melting process needs to consider the latent heat of phase change L , using an equivalent specific heat capacity c_p^* instead of the mass specific heat capacity c_p , as expressed in Eq. (15).

$$c_p^* = c_p + L \frac{df_L}{dT} \quad (15)$$

Glass, as an amorphous material, does not have a fixed melting point. Its viscosity gradually decreases under laser heating, allowing it to flow under the influence of driving forces. The interface between glass and metal is tracked using the level-set variable ϕ , where $\phi = 1$ indicates the glass region, $\phi = 0$ indicates the metal region, and $\phi = 0.5$ is considered the glass-metal interface. The motion of the interface is described by Eq. (16).

$$\frac{\partial \phi}{\partial t} + \mathbf{u} \cdot \nabla \phi = \gamma_i \nabla \cdot \left(\epsilon |\nabla \phi| - \phi (1 - \phi) \cdot \frac{\nabla \phi}{|\nabla \phi|} \right) \quad (16)$$

$\mathbf{u}$ represents the interface velocity, ϵ is the boundary layer thickness, γ_i is the function initialization value.

Heat transfer and fluid flow are described by the Navier-Stokes equations, which consist of the continuity equation Eq. (17), the momentum equation Eq. (18), and the energy equation Eq. (19).

$$\nabla \cdot \mathbf{u} = 0 \quad (17)$$

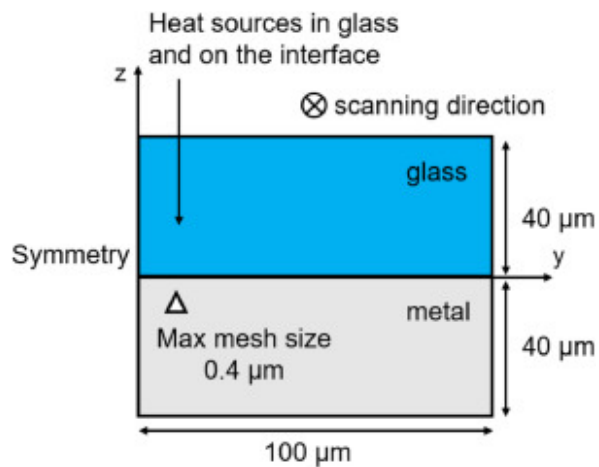
$$\rho \left(\frac{\partial \mathbf{u}}{\partial t} + \mathbf{u} \cdot \nabla \mathbf{u} \right) = -\nabla p + \mu \nabla^2 \mathbf{u} + \mathbf{F}_v \quad (18)$$

$$\rho c_p^* \left(\frac{\partial T}{\partial t} + \nabla \cdot (\mathbf{u} T) \right) = \nabla \cdot (k \nabla T) + S_E \quad (19)$$

where ρ , t , $\mathbf{u}$, p , μ and k are the density, time, vector of velocity, pressure, viscosity and thermal conductivity, respectively. F_v is the volume force, including buoyancy and gravity, and S_E is the heat source term.

2.3.2. Finite element model and boundary conditions

A 2D transient model is established using COMSOL Multiphysics. Due to symmetry, only half is simulated, with the left side being a symmetric boundary, as shown in Fig. 4. The calculation domain is a cross-section perpendicular to the laser scanning direction. To reduce the computational load, the calculation domain is limited to the vicinity of the interface, with both the glass and metal domains having a thickness of 40 μm and a width of 100 μm . The grid size is set to 0.4 μm .



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Fig. 4. Geometry of the level-set two-phase flow model.

When the free electron density in the glass reaches the critical value N_{cr} , the temperature field T_i and the single-pulse heat source Q_i at this moment are output through the multi-pulse temperature field simulation. T_i serves as the initial temperature field value for the flow field simulation. For points on the calculation section, the heat source caused by multiple pulses is considered to be a Gaussian distribution in time, given by Eq. (20).

$$S_E = Q_i f \times \exp\left(-\frac{2t^2}{(\omega_0/v)^2}\right) \quad (20)$$

where f is the laser repetition frequency, v is the laser scanning velocity, and ω_0/v is the time it takes for the laser to traverse the pulse beam waist radius.

At the interface between glass and metal, surface tension, Marangoni force, and the recoil pressure after metal vaporization are considered. Surface tension P_γ acts perpendicular to the interface and is calculated by Eq. (21) and (22) [30].

$$P_\gamma = \kappa \gamma \quad (21)$$

$$\gamma = \gamma_0 + A(T - T_m) \quad (22)$$

where κ is the free surface curvature and γ is the surface tension coefficient, γ_0 is the surface tension at the melting point T_m and A is the temperature coefficient of surface tension.

The Marangoni force P_m acts tangentially to the interface, causing fluid to flow from areas of higher surface tension to areas of lower surface tension, which is described as Eq. (23) [31].

$$P_m = A \nabla_s T \quad (23)$$

where $\nabla_s T$ is the temperature gradient along the surface. The recoil pressure P_r generated by metal vaporization is described as Eq. (24) [32].

$$P_r = P_0 \times \exp\left(\frac{M_a L_v}{R_g} \left(\frac{1}{T_v} - \frac{1}{T}\right)\right) \quad (24)$$

where P_0 is atmospheric pressure, M_a is the molar mass, R_g is the ideal gas constant, L_v is the evaporation latent heat and T_v is the vaporization temperature.

To facilitate comparison with the experiment, the simulation uses the same materials and laser parameters as those in the experiment. The materials chosen are K9 glass and tin, and the laser used is a 400 kHz picosecond laser. The experimental setup consists of a picosecond laser and a galvanometer system, which is specifically described in reference [33]. The specific parameter values can be found in [Table 1](#).

Table 1. The relevant materials and laser parameters.

Name	Symbol	Value	Unit
General Parameters			
Convective heat transfer coefficient	h_0	20	W/m ² K
Interfacial heat transfer coefficient	h	2.3e4	W/m ² K
Time constant for interfacial heat distribution	t_0	2.5e-4	s
Level-set boundary layer thickness	ϵ	0.2	μm
Level-set function initialization value	γ_i	1	m/s
Laser Parameters			
Laser wavelength	λ	1030	nm
Beam waist radius	ω_0	10	μm
Pulse duration	t_p	8	ps
Repetition frequency	f	400	kHz
Laser scanning velocity	v	20	mm/s
Critical free electron density	N_{cr}	1e27	1/m ³
Material Parameters - K9 glass			
Avalanche ionization coefficient [34]	a	0.0017	m ² /J

Name	Symbol	Value	Unit
Multiphoton ionization coefficient [34]	δ_N	1.26e-28	$\text{m}^5/\text{W}^4\text{s}$
Multiphoton ionization order	N	4	
Valence band electron density [35]	n_v	1e28	$1/\text{m}^3$
Electron relaxation time [36]	τ_1	4.3	ps
Damping time [37]	τ	1	fs
Band gap energy [38]	U_i	4	eV
Glass refractive index	n	1.516	
Glass specific heat capacity	c_{p1}	858	J/kgK
Glass thermal conductivity [39]	k_1	$0.818+7.9\text{e-}4\cdot T$ (273.15 K < T < 4000 K)	W/mK
Glass density	ρ_1	2510	kg/m^3
Glass viscosity	μ_1	4e6 (719 °C) 1e3 (1000 °C) 10 (1300 °C)	kg/ms
Glass surface tension	γ_{01}	0.3	N/m
Glass temperature coefficient of surface tension	A_1	-6e-5	N/mK
Material Parameters - Tin [40,41]			
Metal surface reflectivity	R	0.6	
Metal specific heat capacity	c_{p2}	250	J/kgK
Metal thermal conductivity	k_2	67	W/mK
Metal density	ρ_2	7280	kg/m^3

Name	Symbol	Value	Unit
Metel solidus temperature	T_s	490	K
Metel liquidus temperature	T_l	520	K
Metal vaporization temperature	T_v	2533	K
Latent heat of melting	L	4.94e5	J/kg
Latent heat of evaporation	L_v	2.36e6	J/kg
Liquid metal viscosity	μ_2	1.3e-3	kg/ms
Metal surface tension	γ_{02}	0.5	N/m
Metal temperature coefficient of surface tension	A_2	-1e-4	N/mK

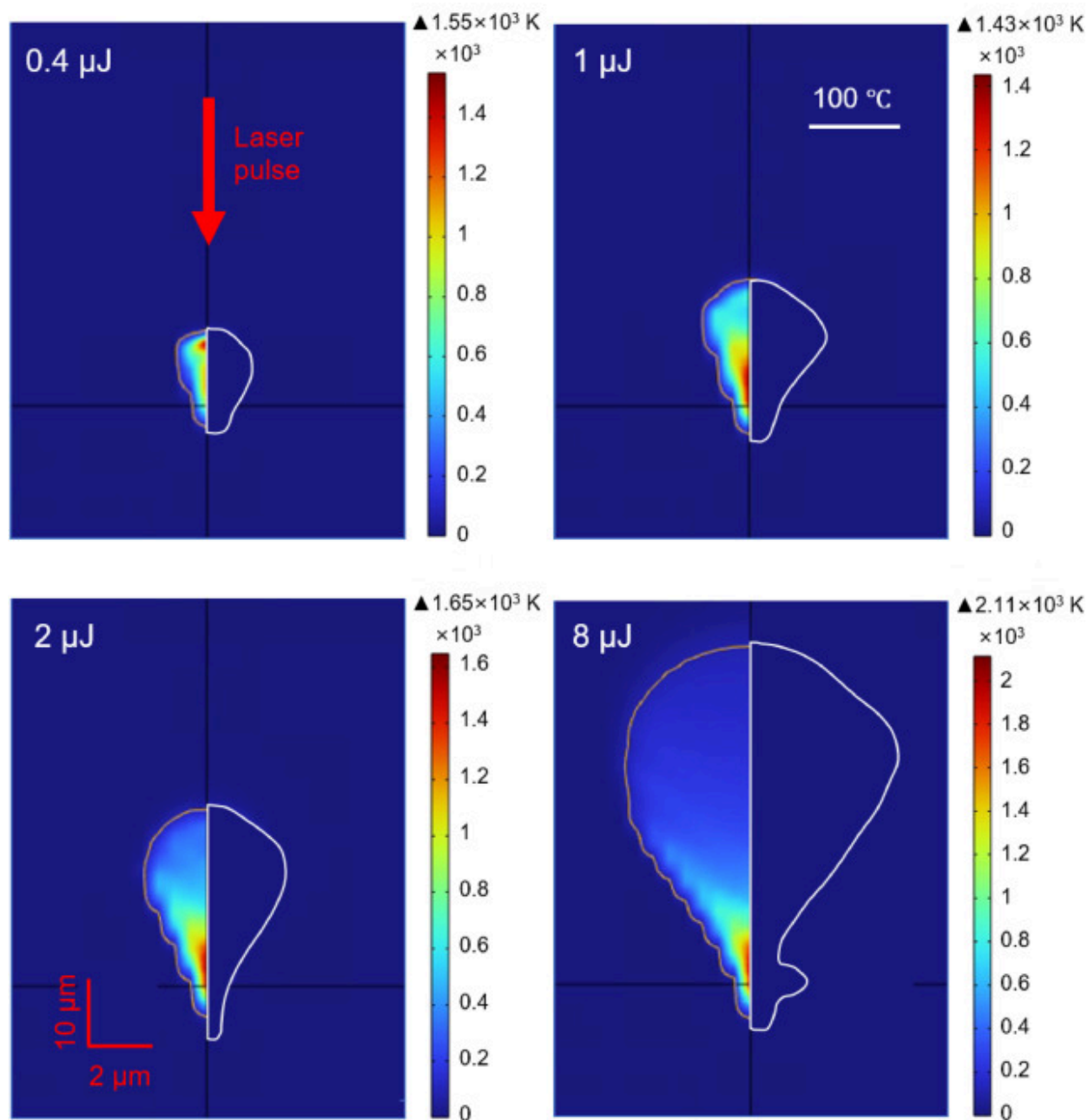
3. Results and discussion

3.1. Validation of the single-pulse model

The 2D steady-state single-pulse model in this study is calculated with the laser parameters from reference [18] to simulate the temperature rise of fused silica under the action of a single femtosecond pulse, with relevant parameters as shown in Table 2. The material parameters of fused silica here differ from those of K9 glass listed in Table 1, requiring adjustments per the specifications in reference [18]. By comparing the results with those from the 3D transient model in the literature, the accuracy of the 2D steady-state model is verified. Fig. 5 shows the single-pulse temperature field under pulse energies ranging from 0.4 μJ to 8 μJ , with the red arrow indicating the laser incidence direction from top to bottom. The left half of the image shows the temperature field results of 2D steady-state single-pulse model, with the white solid line representing the 100 °C isotherm, and the right half shows the corresponding isotherm contours from reference [18]. It can be seen that the temperature field morphology of the present model is close to that in the literature. For a pulse energy of 1 μJ , the maximum temperature rise calculated by the present model is 1430 K, which falls between 800 K [18] and 2000 K [17] in the literature. The comparison of relevant results is shown in Table 3.

Table 2. The relevant laser parameters in reference [18].

Name	Symbol	Value	Unit
Laser wavelength	λ_{re}	800	nm
Pulse energy	E_{re}	0.4–8	μJ
Pulse duration	t_{p_re}	400	fs
Beam waist radius	ω_{0_re}	1	μm



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Fig. 5. Comparison of single-pulse temperature fields within the pulse energy range of 0.4 μJ to 8 μJ : Model in this study (Left) vs. Literature [18] (Right) with 100 °C isotherm highlighted in white.

Table 3. Comparison of simulation results.

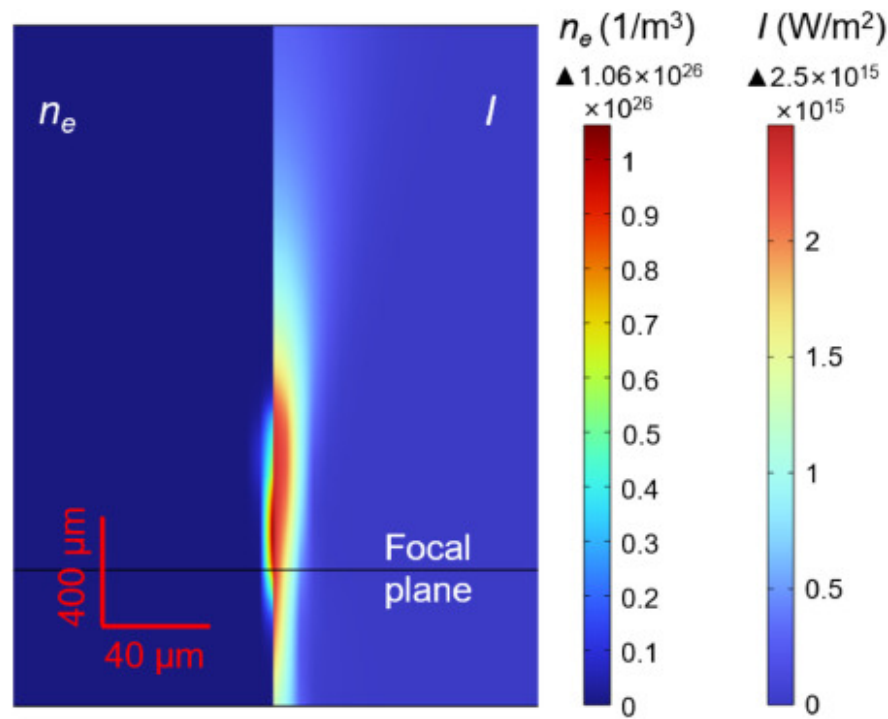
Characteristic quantities	Results in present model	Results in the literature
The maximum temperature rise	1430 K	800 K [18] 2000 K [17]
Size of energy deposition area	3.2 μm $\times$ 24.6 μm	4.8 μm $\times$ 25.7 μm [18] 6.4 μm $\times$ 45.1 μm [17]

The laser energy is mainly deposited near the focus, and the deposition area gradually expands above the focal plane, overall presenting an inverted teardrop shape. The formation process can be simply described as follows. Due to the maximum light intensity at the laser focus, a large amount of laser energy is absorbed by the ionization of the glass here. The plasma generated at the focus will hinder the further propagation of the laser downward, so the area below the focus has less laser energy deposition. Due to the shielding effect of the plasma at the focus, the subsequent laser energy input will be absorbed in the area above the focus, and the laser beam diameter is larger but the light intensity is smaller away from the focus, which makes the laser energy deposition range larger but the amount smaller in that part. As the pulse energy increases from 0.4 μJ to 8 μJ , the range of laser energy deposition gradually increases, and the deposition area gradually extends against the direction of laser incidence.

3.2. Results of the single-pulse model

The distribution of free electron density and light intensity within the glass at an 8 μJ pulse energy is displayed in Fig. 6. Due to symmetry, the left half of the figure shows the distribution of free electron density, while the right half shows the distribution of light intensity. The maximum value of free electron density is $1.06\text{e}26 \text{ 1/m}^3$, which does not reach the critical free electron density, and the maximum value of light intensity is $2.5\text{e}15 \text{ W/m}^2$. The distribution pattern generally takes on an elongated spindle shape, which differs from the inverted teardrop shape shown in Fig. 5. This discrepancy arises because the laser beam

waist radius here is 10 μm , resulting in a larger Rayleigh length, and thus the diameter of the laser beam increases only slightly in areas away from the focal plane.



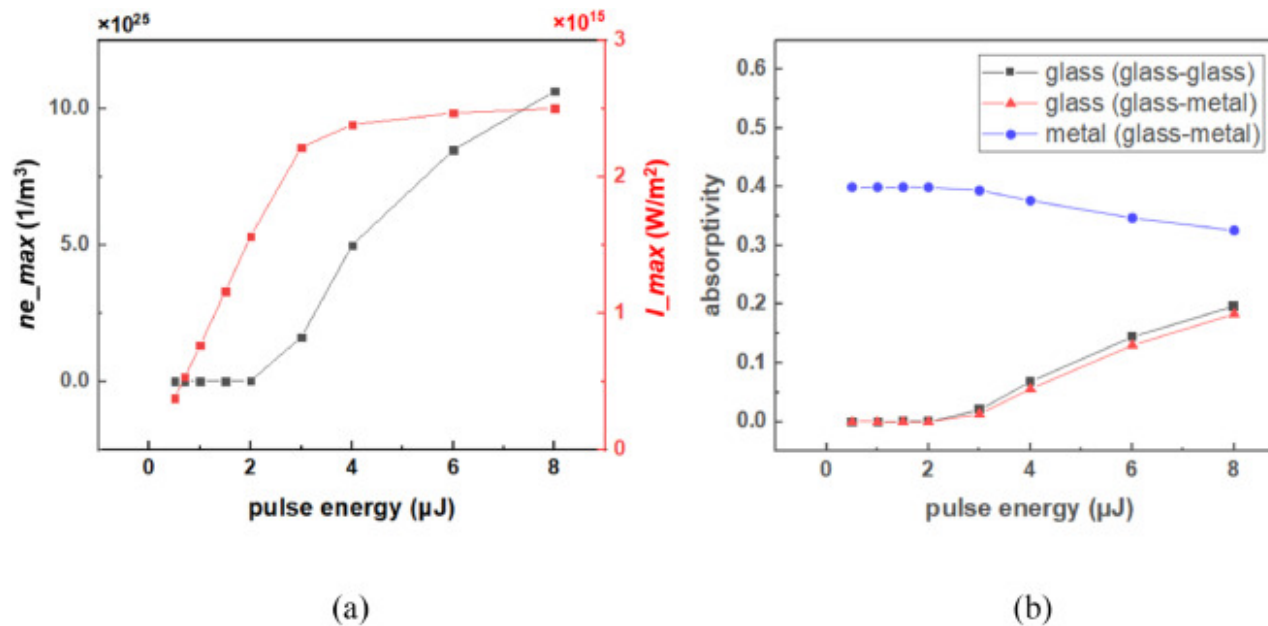
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Fig. 6. Distributions of free electron density and light intensity in glass at a pulse energy of 8 μJ .

Fig. 7(a) illustrates the maximum free electron density (n_{e_max}) and the maximum light intensity (I_{max}) within the glass as the pulse energy increases from 0.5 μJ to 8 μJ . The black line shows the variation of the maximum free electron density; n_{e_max} is relatively low when the pulse energy is $< 2 \mu\text{J}$, but it increases rapidly when the pulse energy exceeds 2 μJ . This indicates that ultrafast lasers exhibit a threshold effect, where ionization in glass is minimal at lower light intensities (for example, a light intensity of $1.5 \times 10^{15} \text{ W/m}^2$ at a pulse energy of 2 μJ), but substantial ionization occurs once the light intensity surpasses the threshold. The red line shows the change in the maximum light intensity; I_{max} increases rapidly when the pulse energy is $< 3 \mu\text{J}$, and it tends to saturate at a value of $2.5 \times 10^{15} \text{ W/m}^2$ when the pulse energy is greater than 3 μJ . This suggests that there is a

clamping effect [42,43] on the light intensity within the glass; a high light intensity can ionize a large amount of plasma within the glass, and the plasma absorbs laser energy, which in turn suppresses the increase in light intensity, causing the light intensity to gradually approach saturation.



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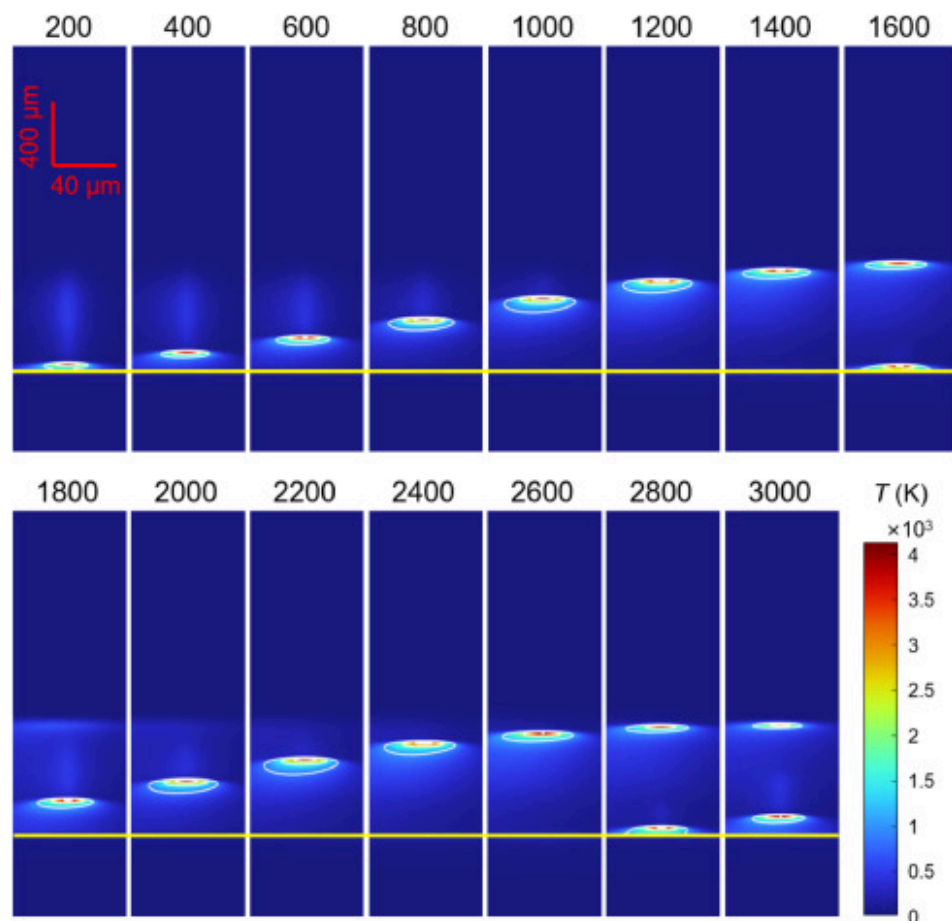
Fig. 7. (a) The maximum free electron density (ne_{max}), the maximum light intensity (I_{max}) within the glass and (b) the laser absorptivity of the material as the pulse energy increases from $0.5 \mu J$ to $8 \mu J$.

Fig. 7(b) shows the absorption of laser energy by the material as the pulse energy increases from $0.5 \mu J$ to $8 \mu J$. The black line indicates the absorption rate of the laser single-pulse energy by the glass in glass welding; the absorption rate is almost zero when the pulse energy is $<2 \mu J$, and it begins to increase when the pulse energy is greater than $2 \mu J$, following the overall trend of ne_{max} in Fig. 7(a). When the pulse energy is $8 \mu J$, the absorption rate reaches its maximum value of 20 %. The red and blue lines show the absorption rates of the laser single-pulse energy by the glass and metal, respectively, in glass-metal welding. As the pulse energy increases, the absorption rate of the glass gradually increases, while that of the metal gradually decreases, but most of the laser energy is still absorbed by the metal. When the pulse energy increases, more plasma is ionized in the glass,

enhancing its laser energy absorption. Since the laser must first transmit through the glass to reach the metal surface, the increased energy absorption by the glass consequently reduces the laser energy arriving at the metal surface. This results in a decrease of the actual energy absorbed by the metal. By comparing the black and red lines, it is observed that in glass welding, the glass's absorption of the laser single-pulse energy is mostly concentrated above the focal plane. This is because the plasma generated at the focus hinders the further propagation of the laser downward, resulting in less laser energy deposition in the area below the focus.

3.3. Results of the multi-pulse temperature field model

The evolution of the multi-pulse temperature field under a pulse energy of 8 μJ and a scanning speed of 20 mm/s is shown in [Fig. 8](#). The numbers at the top of the image represent the number of pulses input, the yellow line represents the glass-metal interface, which is also the focal plane of the laser. The white contour lines are isotherms at 1000 °C, 2000 °C, and 3600 °C. 1000 °C is the working temperature of K9 glass, at which it is considered to have sufficient fluidity and to be in a molten state. 3600 °C is the temperature at which thermo-ionization is significant, where plasma extensively ionizes and is the direct deposition area of laser energy. It can be observed that the laser energy first starts to deposit at the glass-metal interface, which is consistent with the absorption rate in [Fig. 7](#) for a single pulse. The metal surface absorbs more laser energy compared to the inside of the glass, resulting in more noticeable heat accumulation. As pulses continue to be input, the area of laser energy deposition gradually moves upward. This is due to the heat accumulation from previous pulses significantly increasing thermo-ionization, creating a shielding effect for subsequent pulses, causing the glass to absorb the laser higher up. After 1000 pulses, the rate at which the area of laser energy deposition moves upward slows down and eventually stops, then it begins to deposit again at the glass-metal interface, forming a cycle. This is because the laser defocusing is significant, and the laser energy density is not sufficient to maintain heat accumulation, leading to a sudden drop in thermo-ionization, and ultimately failing to maintain the shielding effect.



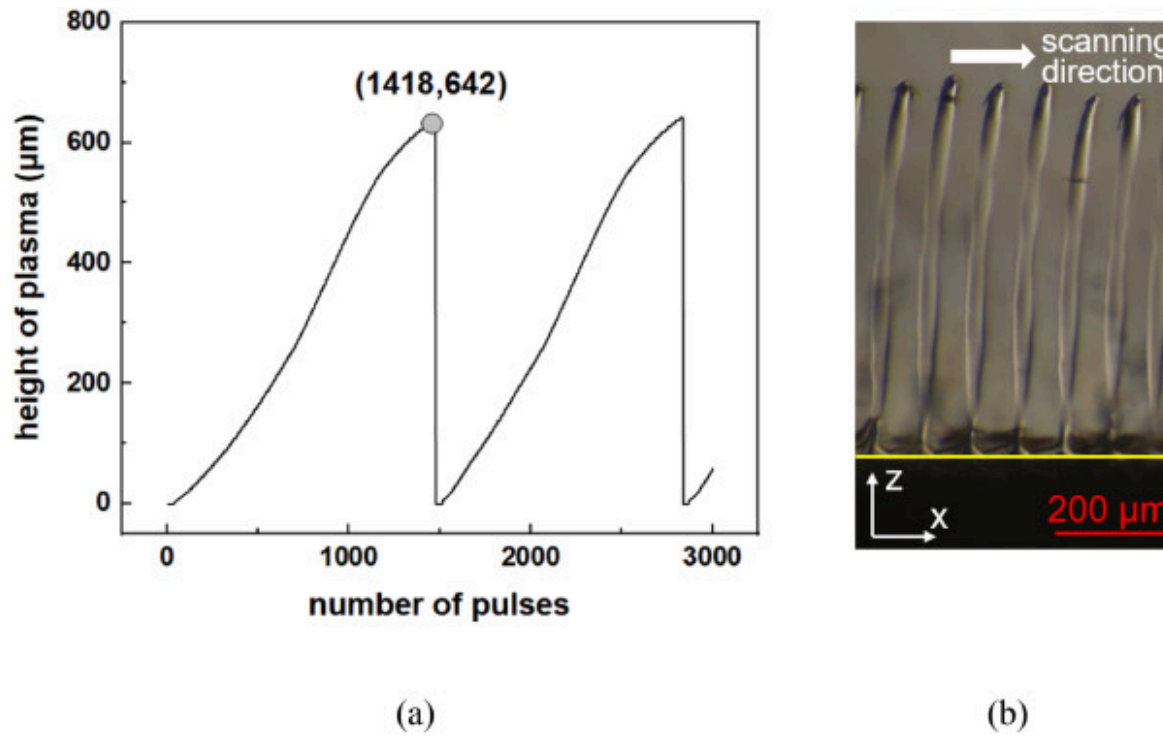
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Fig. 8. Multi-pulse temperature fields for varying pulse numbers at a pulse energy of $8 \mu\text{J}$ and scanning speed of 20 mm/s . The yellow line denotes the glass-metal interface and the white contour lines are isotherms at 1000°C , 2000°C , and 3600°C .

Fig. 9(a) shows the distance between the point of maximum temperature and the glass-metal interface as the number of pulses increases, reflecting the location of the laser energy deposition area. Since the glass absorbs laser energy through ionization, the position of this deposition area also represents the height of the plasma within the glass. Observations of the multi-pulse action process using a high-speed camera reveal that the upward growth of plasma within the glass forms dendritic modified structures

inside the glass [23]. The corresponding experimental result is shown in Fig. 9(b), which uses the same laser parameters as the simulation, namely a pulse energy of 8 μJ and a scanning speed of 20 mm/s. The sample here is observed directly from the side of the glass through a microscope, because the glass has been polished on the side before welding to enhance visualization of internal modified structures. The laser scans along the x-direction, and the yellow line represents the glass-metal interface. The periodic arrangement of dendritic structures inside the glass can be seen, with a height h of 666 μm and a spacing d of 79 μm , meaning the periodic pulse number n calculated as $n = df/v$ is 1580, where f is the laser repetition rate and v is the laser scanning speed. This is relatively consistent with the simulated height of 642 μm and periodic pulse number of 1418. The error of the height h is 3.6 %, and the error of the spacing d is 10.3 %.

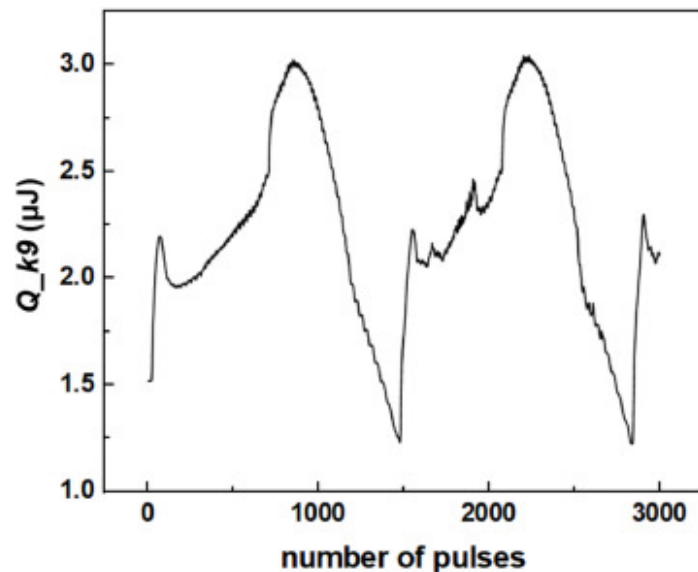


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Fig. 9. Comparison of characteristic quantities of periodic dendritic structures formed by plasma upward motion between (a) simulation result and (b) experimental result. The yellow line marks the glass-metal interface.

Fig. 10 shows the energy absorbed by the glass from each laser pulse as the number of pulses increases. Initially, the energy absorbed by the glass is relatively low; for a laser pulse energy of $8\text{ }\mu\text{J}$, the glass only absorbs $1.5\text{ }\mu\text{J}$. Subsequently, the absorption increases rapidly, peaking at $2.2\text{ }\mu\text{J}$ at around 70 pulses, then drops sharply. Afterward, the absorption rises slowly to a peak of $3.0\text{ }\mu\text{J}$ at around 850 pulses before gradually decreasing again, completing a cycle by 1418 pulses. The two processes of rising and falling absorption may be related to different modes of laser energy deposition. The first rise and fall occur when the laser energy is deposited near the glass-metal interface, where the laser intensity is high enough to produce a large number of seed electrons through multi-photon ionization, so the energy deposition is dominated by light intensity at this time. The second rise and fall occur when the laser energy is deposited inside the glass, away from the glass-metal interface, where the laser intensity is weaker, but the temperature is still high enough to produce sufficient seed electrons through thermo-ionization, so the energy deposition is dominated by temperature at this time.



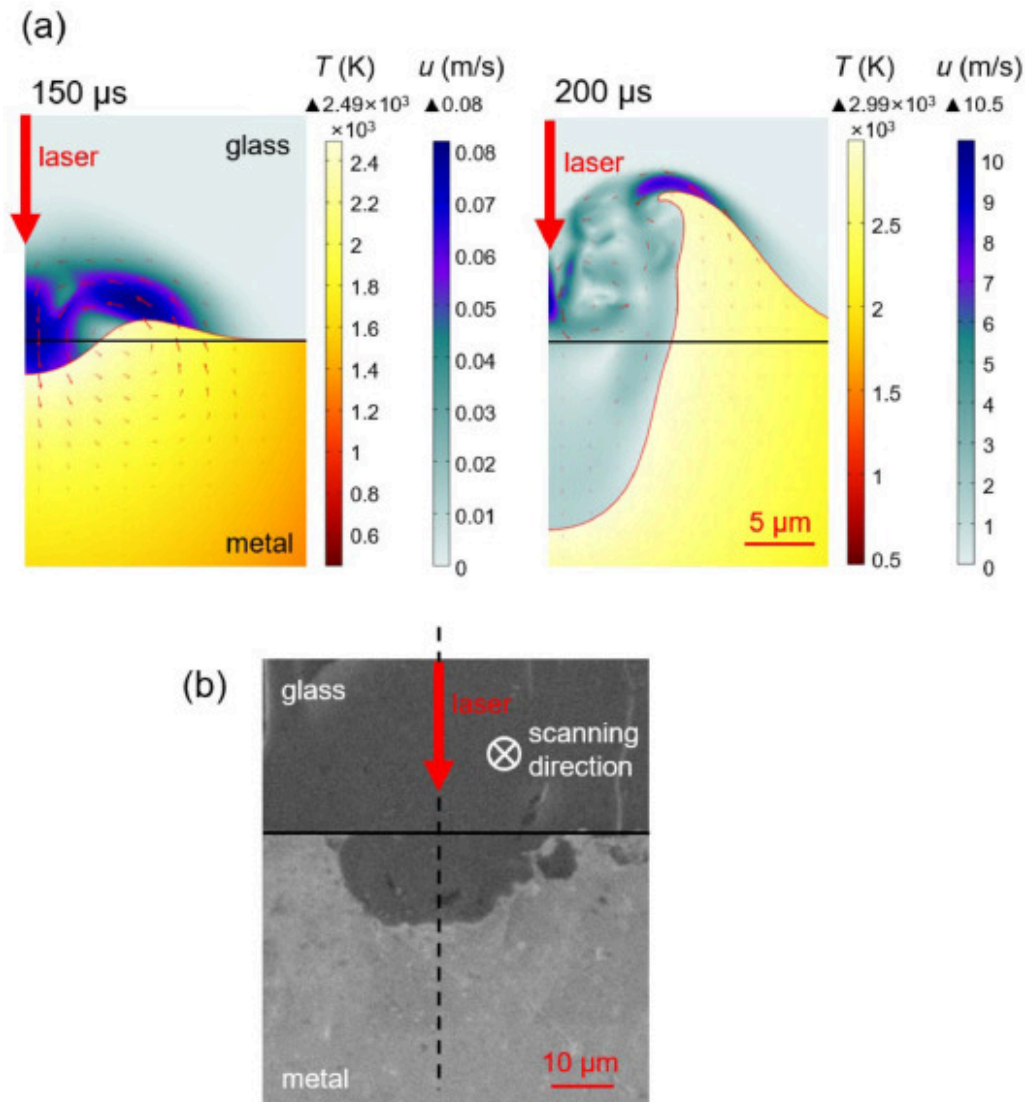
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Fig. 10. The energy absorbed by the glass from each laser pulse as the number of pulses increases.

3.4. Results of the multi-pulse flow field model

Fig. 11(a) shows the simulation results of the flow field at the interface. The black horizontal line represents the initial glass-metal interface, with the glass above and the metal below. The red line represents the current glass-metal interface. The velocity field of the glass domain is displayed in blue, and the temperature field in the metal domain is shown in yellow. The arrow arrays in the figure represent the magnitude and direction of the velocity. It can be seen that at 150 μs , the center of the metal is indented downward by 2.5 μm , and the edges are bulging upward by 1.4 μm . The velocity field generally forms a vortex, with a maximum value of 0.08 m/s, and the velocity direction at the interface is essentially perpendicular to the interface. This indicates that it is caused by the recoil pressure from metal vaporization. At 200 μs , the center of the metal is indented downward by 13.4 μm , and the edges are bulging upward by 10.7 μm ; the glass embeds more deeply into the metal. The maximum velocity at this time is 10.5 m/s, located at the edge where the metal is bulging upward.



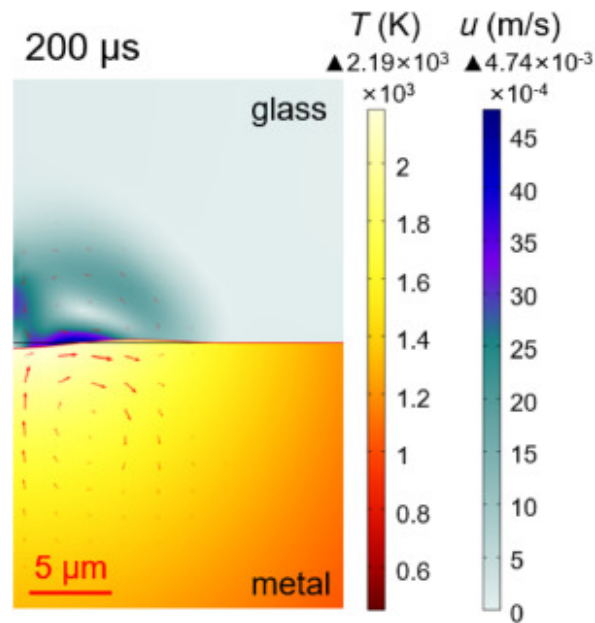
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Fig. 11. (a) Simulation results of the flow field at the interface, where the velocity field of the glass domain is displayed in blue, and the temperature field in the metal domain is shown in yellow. (b) Cross-sectional SEM micrograph of the weld seam. The black horizontal line represents the initial glass-metal interface.

Comparing with the experimental result in Fig. 11(b), which is observed through SEM on the weld section obtained by grinding and polishing, the center of the metal is indented downward by $14.7\text{ }\mu\text{m}$, but the edges show almost no bulging. This suggests that the material from the center of the metal did not transfer to the edges to form a pile-up but was directly removed, possibly because there was still a certain gap between the glass and the metal during the welding process, and the vaporized metal escaped through the gap. The absence of a pile-up at the edges also results in a wider width of the metal pit.

Fig. 12 presents the simulation results of the flow field at the interface without considering the vaporization recoil pressure. It can be observed that the depth of the metal center's indentation is extremely shallow, only $0.3\text{ }\mu\text{m}$. The velocity field shows a vortex on both the glass and metal sides, with a maximum value of $4.74\text{e-}3\text{ m/s}$. The velocity direction at the interface is essentially parallel to the interface, indicating that it is primarily driven by Marangoni forces. Therefore, the vaporization recoil pressure is crucial for the interlocking of glass and metal.



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Fig. 12. The simulation results of the flow field at the interface without considering the recoil pressure.

4. Conclusions

A thermo-fluid coupled model spanning from single-pulse to multi-pulse is established. By dimensionality reduction of the single-pulse model to a two-dimensional steady-state, computational efficiency is significantly enhanced, enabling the solution for each single pulse within the multi-pulse model to demonstrate the variance in laser energy deposition. This approach successfully simulates the uneven energy deposition during the high-repetition rate ultrafast laser welding of glass and metal. The simulation results correlate well with experimental observations, confirming the model's validity and precision. Analysis through the model reveals that in the high-repetition rate laser welding process, the deposition of laser energy is notably influenced by thermo-ionization, leading to a periodic shift of the laser energy deposition area from the glass-metal interface into the glass interior. Furthermore, the model indicates that the flow behavior at the glass and metal interface during welding is jointly affected by surface tension, Marangoni forces, and metal vaporization recoil pressure, with the recoil pressure being the predominant driving force. The findings of this study offer crucial theoretical support and guidance for the advancement of high-repetition rate ultrafast laser welding technology.

CRedit authorship contribution statement

Chenyun Tian: Writing – original draft, Visualization, Validation, Investigation, Formal analysis, Data curation. **Hong Shen:** Writing – review & editing, Supervision, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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